

Monge Formulation

Exercise statements

This version contains the exercises only. Hints and worked solutions are available in the companion PDF.

Exercise 1 (Discrete assignment).

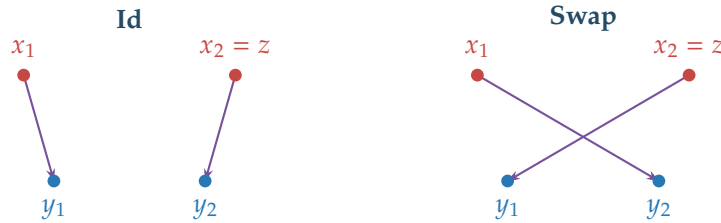
Let $n = 3$ and $C = \begin{pmatrix} 2 & 1 & 3 \\ 3 & 2 & 2 \\ 4 & 3 & 1 \end{pmatrix}$. Identify every optimal assignment.

Exercise 2 (Two points to two points: when does each permutation win?).

Fix $x_1, y_1, y_2 \in \mathbb{R}^d$ with $y_1 \neq y_2$ and let $x_2 = z \in \mathbb{R}^d$ vary. With quadratic cost $c(x, y) = \|x - y\|^2$, compare the two assignments

$$\text{Id: } x_1 \mapsto y_1, z \mapsto y_2 \quad \text{vs} \quad \text{Swap: } x_1 \mapsto y_2, z \mapsto y_1.$$

Show that the set of z where Id is no more expensive than Swap is a half-space bounded by a hyperplane orthogonal to $y_1 - y_2$. Give the equation of this hyperplane and explain that it divides $\mathbb{R}^d$ into two regions, one for each optimal permutation.



Exercise 3 (Discrete $\rightarrow$ continuous and continuous $\rightarrow$ discrete).

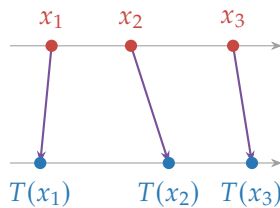
(1) In 1D let $\alpha = \sum_{i=1}^N a_i \delta_{x_i}$ (finite support) and let β be absolutely continuous. Show that there is no measurable map T with $T_{\#}\alpha = \beta$.

(2) Conversely, let α be absolutely continuous on $\mathbb{R}$ and let $\beta = \sum_{j=1}^M b_j \delta_{y_j}$ with $\sum_j b_j = 1$. Build maps T (many, up to α -a.e. equality, if β has at least two distinct atoms of positive mass) with $T_{\#}\alpha = \beta$.

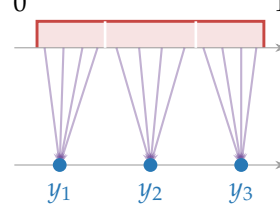
(3) Let $\alpha = \mathcal{U}[0, 1]$ and let $\beta = \sum_{j=1}^M b_j \delta_{y_j}$ on $\mathbb{R}$ with $b_j > 0$, $\sum_j b_j = 1$, and $y_1 < \dots < y_M$. For $p \geq 1$, compute an $\mathcal{W}_p$ -optimal Monge map $T : [0, 1] \rightarrow \mathbb{R}$ from α to β .

(4) Suppose $b_j = 1/n$ for $j = 1, \dots, n$ and $y_j = \frac{j-1/2}{n}$ (midpoints of the n equal subintervals of $[0, 1]$). Compute $W_2^2(\alpha, \beta)$.

A map sends atoms to atoms



An interval can collapse to an atom



Exercise 4 (Uniform laws on intervals, quadratic cost).

Let $\alpha = \mathcal{U}_{[a,b]}$ and $\beta = \mathcal{U}_{[c,d]}$ on $\mathbb{R}$ with $a < b$, $c < d$, and $c(x, y) = |x - y|^2$.

- (i) Find the Monge optimal transport map T .
- (ii) Compute $\mathcal{W}_2^2(\alpha, \beta)$ in closed form.

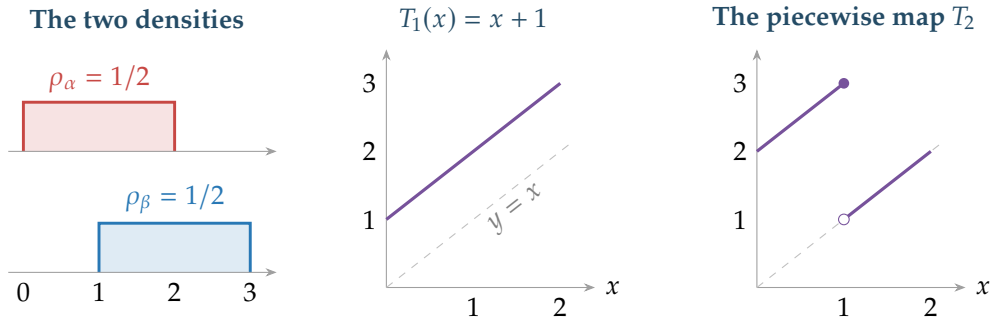
Exercise 5 (Many Monge solutions for $|x - y|$).

Let $\alpha = \mathcal{U}_{[0,2]}$, $\beta = \mathcal{U}_{[1,3]}$ and $c(x, y) = |x - y|$.

- (i) Show $T_1(x) = x + 1$ is optimal and $\mathcal{W}_1(\alpha, \beta) = 1$.
- (ii) Show

$$T_2(x) = \begin{cases} x + 2, & x \in [0, 1], \\ x, & x \in (1, 2], \end{cases}$$

is also optimal.

**Exercise 6** (Affine images).

(1) **1-D case.** Let $\mu \in \mathcal{P}_2(\mathbb{R})$ and $\nu = (x \mapsto ax + b)_\# \mu$ with $a > 0$. What is the Monge map from μ to ν for the quadratic cost? Compute the W_2 distance in terms of the mean $m_\mu = \int x d\mu$ and variance $\text{Var}(\mu) = \int (x - m_\mu)^2 d\mu$.

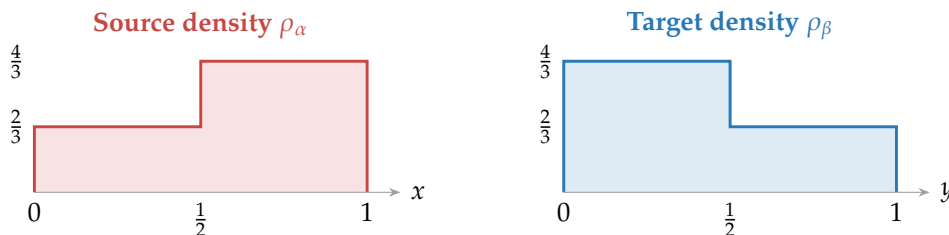
(2) **Higher dimension.** Let $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ and $\nu = (x \mapsto Ax + b)_\# \mu$ with $A \in \mathbb{R}^{d \times d}$. Give a simple condition on A ensuring that $T(x) = Ax + b$ is the Brenier map (assume μ is absolutely continuous). Under this condition, compute the W_2 distance in terms of $m_\mu = \mathbb{E}[X]$ and the covariance $\Sigma_\mu = \mathbb{E}[(X - m_\mu)(X - m_\mu)^\top]$.

Exercise 7 (Piecewise constant densities on $[0, 1]$, quadratic cost).

Let α and β have densities

$$\rho_\alpha(x) = \begin{cases} \frac{2}{3}, & 0 \leq x < \frac{1}{2}, \\ \frac{4}{3}, & \frac{1}{2} \leq x \leq 1, \end{cases} \quad \rho_\beta(x) = \begin{cases} \frac{4}{3}, & 0 \leq x < \frac{1}{2}, \\ \frac{2}{3}, & \frac{1}{2} \leq x \leq 1, \end{cases}$$

and zero outside $[0, 1]$. Construct and sketch the optimal Monge map for the quadratic cost, and compute $\mathcal{W}_2^2(\alpha, \beta)$.



Exercise 8 (Translation decomposition for $\mathcal{W}_2$).

Let $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$ and assume that α has a Lebesgue density. Throughout this exercise, use the Monge formulation

$$\mathcal{W}_2^2(\alpha, \beta) = \inf_{S_{\#}\alpha = \beta} \int \|x - S(x)\|^2 d\alpha(x).$$

For $u \in \mathbb{R}^d$ write $T_u(x) = x + u$.

- (a) Show there exist unique vectors a, b such that the *centred* measures

$$\alpha_0 := (T_{-a})_{\#}\alpha, \quad \beta_0 := (T_{-b})_{\#}\beta$$

satisfy $\int x d\alpha_0(x) = \int x d\beta_0(x) = 0$, and

$$\alpha = (T_a)_{\#}\alpha_0, \quad \beta = (T_b)_{\#}\beta_0.$$

- (b) Prove the decomposition

$$\boxed{\mathcal{W}_2^2(\alpha, \beta) = \|a - b\|^2 + \mathcal{W}_2^2(\alpha_0, \beta_0)}.$$

- (c) If T_0 is the Brenier map from α_0 to β_0 , show that

$$T(x) := T_0(x - a) + b$$

is the Brenier map from α to β .

Exercise 9 (Adding separable terms; quadratic cost vs. $-\langle x, y \rangle$).

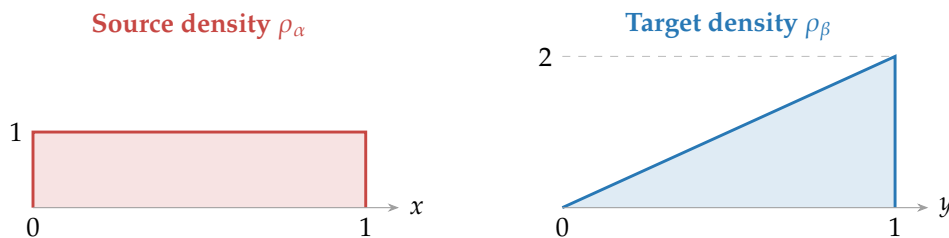
Fix $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$ and consider Monge maps T with $T_{\#}\alpha = \beta$.

- (a) Let $c : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ be a measurable cost and define $\tilde{c}(x, y) = c(x, y) + u(x) + v(y)$, with $u : \mathbb{R}^d \rightarrow \mathbb{R}, v : \mathbb{R}^d \rightarrow \mathbb{R}$. Assume $u \in L^1(\alpha), v \in L^1(\beta)$, and that the integral of c is well-defined for every admissible map (for instance $c \geq 0$). Show that the sets of Monge minimizers for c and for $\tilde{c}$ (with the same marginals) coincide.
- (b) Let $\lambda > 0$ and define $\hat{c}(x, y) = \lambda c(x, y)$. Show that the Monge minimizers for c and $\hat{c}$ coincide.
- (c) Take $c_{\text{quad}}(x, y) = \|x - y\|^2$, $c_{\text{lin}}(x, y) = -\langle x, y \rangle$. Prove that c_{quad} and c_{lin} have the same Monge minimizers (for fixed α, β).

Exercise 10 (Quantile coupling: explicit computation).

Let $\alpha = \mathcal{U}[0, 1]$ and let β be supported on $[0, 1]$ with density $\rho_{\beta}(y) = 2y$.

- (a) Compute the cumulative functions C_{α}, C_{β} and their quantiles $C_{\alpha}^{-1}, C_{\beta}^{-1}$.
- (b) Using monotone transport in 1D, write the optimal map $T = C_{\beta}^{-1} \circ C_{\alpha}$.
- (c) Compute $\mathcal{W}_2^2(\alpha, \beta)$ using both $\int_0^1 |x - T(x)|^2 dx$ and the quantile formula $\int_0^1 |C_{\alpha}^{-1}(r) - C_{\beta}^{-1}(r)|^2 dr$.



Exercise 11 (Cumulative functions).

Let α be a probability measure on $\mathbb{R}$ and let its cumulative distribution function be

$$C_\alpha(x) = \alpha((-\infty, x]), \quad x \in \mathbb{R}.$$

Consider the pushforward by C_α , namely $(C_\alpha)_\# \alpha$.

- (1) Assume α is atomless (equivalently, C_α is continuous). This includes, but is not restricted to, measures with a Lebesgue density. Prove that

$$(C_\alpha)_\# \alpha = \mathcal{U}[0, 1].$$

In your proof, clearly point out *where* the continuity assumption is used.

- (2) What happens if α is purely discrete, say $\alpha = \sum_i p_i \delta_{x_i}$ with $\sum_i p_i = 1$ and $x_1 < x_2 < \dots$? Describe the law of $C_\alpha(X)$ for $X \sim \alpha$, and show that for all $t \in [0, 1]$,

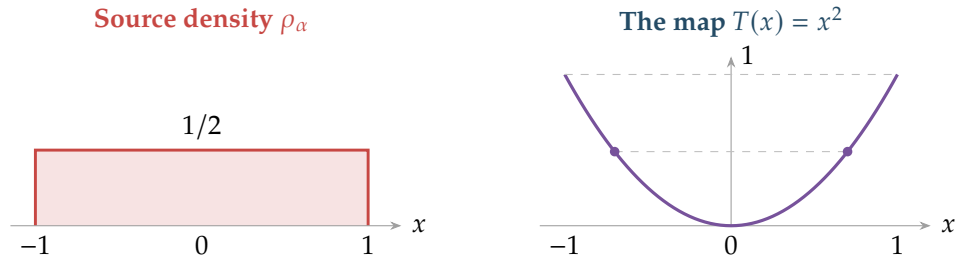
$$\mathbb{P}(C_\alpha(X) \leq t) \leq t,$$

with equality at the jump values of C_α .

Exercise 12 (Pushforward of a uniform law by a nonlinear map).

Let $\alpha = \mathcal{U}[-1, 1]$ and consider the map $T : \mathbb{R} \rightarrow \mathbb{R}$, $T(x) = x^2$.

- (1) Compute the law $\beta = T_\# \alpha$, i.e. find its density with respect to Lebesgue measure.
 (2) Check that this density integrates to 1 over its support.
 (3) Sketch the graph of the density and comment on its behaviour near 0 and 1.

**Further exercises****Exercise 13 (Radial symmetry).**

Let $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$ be invariant under all orthogonal maps, and assume α is absolutely continuous. Define the radial cumulative function $F_\mu(r) = \mu(\{x : \|x\| \leq r\})$, $r \geq 0$, and its generalized inverse for $0 < u < 1$.

- (i) Show that the unique quadratic optimal map commutes with orthogonal maps: $Q \circ T \circ Q^{-1} = T$ α -a.e. Show that it has a radial representative $T(x) = t(\|x\|)x/\|x\|$ for α -a.e. $x \neq 0$, with t nonnegative and nondecreasing on the relevant radii.
 (ii) Prove $t(r) = F_\beta^{-1}(F_\alpha(r))$ for almost every source radius. Does $F_\beta(t(r)) = F_\alpha(r)$ still hold if β has an atom on a sphere?
 (iii) Specialize to uniform laws on balls of positive radii R_α, R_β . Give the map and its squared cost.